220109: sandstone, Wendy 1

(unassigned unit, northern Perth Basin)

Location and sampling

Geraldton (SH 50-1), Mungo (1941)
MGA Zone 50, 305515E 6868070N

Sampled on 24 March 2017

This sample was collected from the 323.12 – 324.12 m depth interval of drillcore Wendy 1, a stratigraphic well drilled in 2004 by Victoria International Petroleum NL (Victoria Petroleum, 2004). The drillhole is located on the eastern side of Dartmoor Road, about 30 km northeast of Mount Erin, 13.4 km west of Wandana Hill, and 3.2 km north of the Yuna–Tenindewa Road.

Geological context

The unit sampled is a quartz sandstone of the northern Perth Basin. Although the siliciclastic interval from 112.0 to 1210.5 m in this drillcore has previously been interpreted as Tumblagooda Sandstone (Victoria Petroleum, 2004; Retallack, 2009), Allen and Trinajstic (2017) considered it to be lithologically distinct from the composite type section of the Tumblagooda Sandstone in the Murchison River gorge section. The presence of well-developed paleosols and preserved rootlets in Wendy 1 is not comparable to outcrop sections of the Tumblagooda Sandstone (Allen and Trinajstic, 2017). The age of this unit in Wendy 1 is constrained by an Early Devonian fish assemblage (Allen and Trinajstic, 2017), and may be coeval with or slightly younger than the Tumblagooda Sandstone. Zircons from a sandstone sample taken from about 784 m depth in Wendy 1, and tentatively assigned to the Tumblagooda Sandstone, yielded an age of 502 ± 3 Ma for the youngest zircon and a conservative maximum depositional age of 524 ± 4 Ma (GSWA 220110, Wingate et al., 2021). A maximum age for deposition of the Tumblagooda Sandstone is provided by a 434 ± 16 Ma Pb-isotope age for Pb mineralization in stratigraphically underlying metamorphic rocks of the Northampton Inlier (Richards et al., 1985). A laser ablation U–Pb detrital zircon study of 16 Tumblagooda Sandstone samples yielded a youngest zircon at 466 ± 8 Ma (1σ) and a conservative maximum depositional age of 481 ± 4 Ma (GSWA 214841, Markwitz et al., 2017). Based on comparable trace fossil assemblages and plant remains (H Allen, 2021, written comm., 20 October) the Tumblagooda Sandstone may be latest Silurian to Early Devonian in age.

Petrographic description

The sample is a homogeneous, pale yellow, quartz-rich sandstone bed about 1 m thick, between grey–green sandstone and dark red claystone (Fig. 1). A detailed petrographic description is not available for this sample. However, petrography of other samples from Wendy 1 is reported by Kettanah et al. (2015, and references within).

Zircon morphology

Zircons isolated from this sample are mainly colourless, anhedral to euhedral, and strongly rounded. The crystals are up to 250 µm long, and equant to elongate, with aspect ratios up to 6:1. Some crystals have pitted outer surfaces and, in cathodoluminescence (CL) images, many exhibit concentric zoning truncated at grain edges, features consistent with abrasion during sedimentary transport. A CL image of representative zircons is shown in Figure 2.

Analytical details

This sample was analysed on 12 June 2018, using LA-ICP-MS instruments at the GeoHistory Facility in the John de Laeter Centre, Curtin University. Separated zircons mounted in epoxy disks were ablated using a Resonetics RESOlution M-50A-LR sampling system, incorporating a 193 nm Compex 102 excimer laser, and ion signal intensities were measured using an Agilent 7700s quadrupole ICP-MS. Samples were ablated for 35 s at a 5 Hz repetition rate, using a 38 µm beam diameter and laser energy of 2.5 J/cm^2. Seventeen analyses of the 91 500 zircon standard obtained during the session indicated a concordia age of 1062.7 ± 2.1 Ma (MSWD = 0.84), in agreement with the accepted value of 1062.4 Ma. Secondary zircon standards GJ1, OGC1 and Plešovice also yielded mean dates in agreement with their accepted values. Additional analytical details are provided in Fielding et al. (2021). All relevant uncertainties have been propagated into the ratios and dates reported in Table 1. Dates for analyses for which $^{238}U/^{206}Pb$ ratios indicate dates <1000 Ma are based on 207-corrected $^{238}U/^{206}Pb^*$ ratios; those >1000 Ma are based on uncorrected $^{207}Pb/^{206}Pb$ ratios.

Results

A total of 150 analyses were obtained from 150 zircons. Results are listed in Table 1, and shown in concordia diagrams (Figs 3, 4) and a probability density plot (Fig. 5).

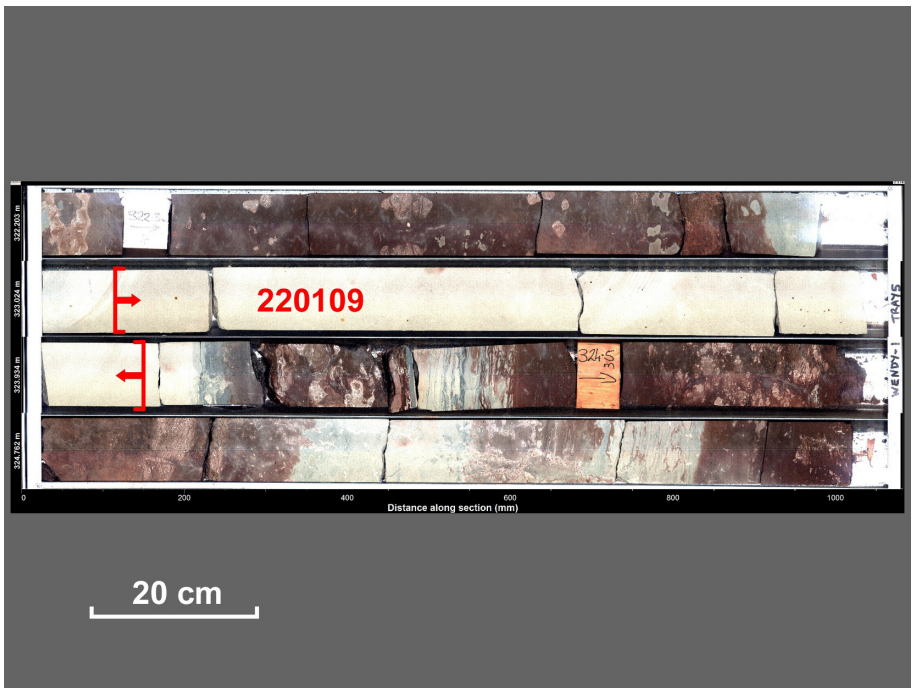

Figure 1. **Drillcore (HyLogger-3) image of sample 220109: sandstone, Wendy 1. The sample interval is highlighted in red**

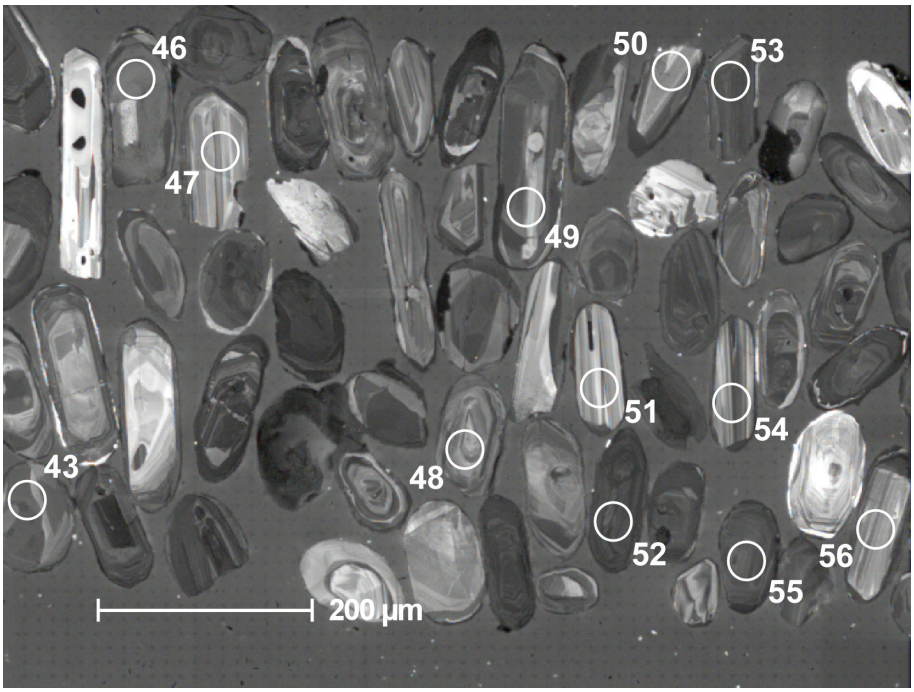

Figure 2. **Cathodoluminescence image of representative zircons from sample 220109: sandstone, Wendy 1. Numbered circles indicate the approximate locations of analysis sites**

Table 1. LA-ICP-MS analytical results for zircons from sample 220109: sandstone, Wendy 1

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb^*$ ± 1σ		$^{207}Pb^*/^{206}Pb^*$ ± 1σ		Disc. (%)
Y	94	94	70	141	2.02	12.330	0.084	0.06690	0.00185	503	3	835	58	39.8
S	118	118	56	77	1.36	12.438	0.085	0.05790	0.00110	499	3	526	42	5.2
S	112	112	106	128	1.21	12.317	0.061	0.06050	0.00120	503	2	622	43	19.0
S	134	134	42	90	2.13	12.107	0.088	0.06080	0.00225	512	4	632	80	19.1
S	91	91	59	115	1.95	12.005	0.079	0.06640	0.00225	516	3	819	71	37.0
S	98	98	423	415	0.98	12.041	0.048	0.05704	0.00050	514	2	493	19	-4.3
S	138	138	87	141	1.63	11.972	0.069	0.06170	0.00105	517	3	664	36	22.1
S	113	113	75	185	2.48	12.063	0.080	0.05540	0.00100	513	3	428	40	-19.8
S	47	47	77	147	1.91	11.915	0.071	0.05660	0.00105	520	3	476	41	-9.2
S	96	96	58	69	1.20	11.862	0.084	0.05900	0.00110	522	4	567	41	8.0
S	6	6	133	200	1.50	11.763	0.053	0.05920	0.00075	526	2	574	28	8.4
S	48	48	157	242	1.54	11.754	0.064	0.05710	0.00075	526	3	495	29	-6.3
S	75	75	144	236	1.64	11.784	0.066	0.05480	0.00065	525	3	404	27	-29.9
S	115	115	64	86	1.35	11.730	0.069	0.05670	0.00130	527	3	480	51	-9.9
S	10	10	188	108	0.58	11.682	0.054	0.05868	0.00060	529	2	555	22	4.6
S	25	25	77	150	1.94	11.561	0.074	0.05930	0.00095	535	3	578	35	7.5
S	84	84	77	217	2.83	11.447	0.066	0.06170	0.00125	540	3	664	43	18.7
S	21	21	228	521	2.29	11.436	0.049	0.05870	0.00065	540	2	556	24	2.8
S	106	106	316	201	0.64	11.410	0.050	0.05780	0.00065	542	2	522	25	-3.7
S	43	43	308	167	0.54	11.330	0.047	0.05870	0.00055	545	2	556	20	1.9
S	126	126	362	70	0.19	11.210	0.046	0.05766	0.00050	551	2	517	19	-6.6
S	108	108	569	779	1.37	11.068	0.042	0.05788	0.00049	558	2	525	19	-6.2
S	35	35	429	44	0.10	10.911	0.043	0.05768	0.00050	565	2	518	19	-9.2
S	2	2	178	62	0.35	10.810	0.044	0.05960	0.00065	570	2	589	24	3.2
S	132	132	58	4	0.06	10.688	0.057	0.05730	0.00095	577	3	503	36	-14.6
S	129	129	205	136	0.66	10.578	0.038	0.05933	0.00060	582	2	579	22	-0.5
S	18	18	287	108	0.38	10.504	0.099	0.06040	0.00060	586	5	618	21	5.1
S	79	79	219	179	0.82	10.142	0.093	0.06000	0.00065	606	5	604	23	-0.4
S	97	97	195	63	0.32	10.106	0.043	0.05990	0.00060	608	2	600	22	-1.4
S	143	143	30	29	0.98	9.268	0.082	0.06290	0.00180	661	6	705	61	6.3
S	46	46	151	75	0.49	9.001	0.049	0.06224	0.00060	679	3	682	21	0.5
S	66	66	95	36	0.38	8.913	0.060	0.06210	0.00080	686	4	678	28	-1.2
S	100	100	67	160	2.38	8.741	0.046	0.06160	0.00100	698	3	660	35	-5.8
S	149	149	60	62	1.03	8.673	0.049	0.06600	0.00115	703	4	806	36	12.8
S	50	50	85	46	0.54	8.673	0.045	0.06480	0.00075	703	3	768	24	8.4
S	44	44	209	153	0.73	8.557	0.037	0.06960	0.00070	713	3	917	21	22.3
S	68	68	557	418	0.75	8.552	0.031	0.06147	0.00047	713	2	656	16	-8.7

Table 1. continued

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	$^{238}U/^{206}Pb$	± 1σ	$^{207}Pb/^{206}Pb$	± 1σ	$^{238}U/^{206}Pb^*$	± 1σ	$^{207}Pb^*/^{206}Pb^*$	± 1σ	Disc. (%)
S	101	101	96	125	1.31	8.511	0.043	0.06330	0.00095	716	3	718	32	0.3
S	27	27	251	282	1.12	8.499	0.036	0.06396	0.00060	717	3	740	20	3.1
S	121	121	85	43	0.50	8.264	0.044	0.06330	0.00085	736	4	718	29	-2.5
S	102	102	110	77	0.71	8.032	0.055	0.07210	0.00100	756	5	989	28	23.5
S	5	5	398	159	0.40	7.782	0.042	0.06670	0.00050	779	4	828	16	5.9
S	122	122	424	181	0.43	7.752	0.039	0.06840	0.00050	782	4	881	15	11.2
S	8	8	254	192	0.76	7.675	0.053	0.06754	0.00055	790	5	854	17	7.6
S	51	51	119	54	0.46	7.610	0.055	0.06540	0.00075	796	5	787	24	-1.1
S	92	92	110	60	0.54	7.564	0.049	0.06490	0.00070	800	5	771	23	-3.8
S	49	49	118	38	0.32	7.380	0.057	0.06600	0.00075	819	6	806	24	-1.6
S	12	12	109	64	0.59	7.294	0.040	0.06760	0.00075	828	4	856	23	3.3
S	11	11	230	140	0.61	7.231	0.039	0.06972	0.00060	835	4	920	18	9.2
S	37	37	237	76	0.32	7.215	0.034	0.06748	0.00055	837	4	853	17	1.9
S	9	9	554	186	0.33	7.200	0.026	0.06719	0.00049	838	3	844	15	0.6
S	40	40	185	51	0.28	7.128	0.033	0.07130	0.00065	846	4	966	19	12.4
S	54	54	171	182	1.06	6.878	0.033	0.06830	0.00070	875	4	878	21	0.3
S	60	60	140	151	1.08	6.618	0.031	0.06930	0.00075	907	4	908	22	0.1
S	31	31	145	49	0.34	6.596	0.033	0.06930	0.00070	910	4	908	21	-0.2
S	85	85	31	18	0.58	6.536	0.051	0.07540	0.00200	918	7	1079	53	15.0
S	26	26	755	76	0.10	6.559	0.024	0.06760	0.00048	915	3	856	15	-6.8
S	128	128	182	195	1.07	6.518	0.025	0.06970	0.00065	920	3	920	19	-0.1
S	72	72	36	32	0.89	6.443	0.046	0.06890	0.00120	930	6	896	36	-3.8
S	133	133	109	82	0.75	6.382	0.039	0.06820	0.00075	938	5	875	23	-7.3
S	103	103	329	183	0.56	6.231	0.025	0.07059	0.00055	960	4	946	16	-1.5
S	33	33	179	84	0.47	6.101	0.028	0.07370	0.00070	978	4	1033	19	5.3
S	73	73	155	1	0.01	6.028	0.031	0.07230	0.00065	989	5	994	18	0.5
S	99	99	63	85	1.35	5.977	0.034	0.07500	0.00090	997	5	1069	24	6.7
S	90	90	147	149	1.01	5.724	0.033	0.07274	0.00060	1038	5	1007	17	-3.1
S	148	148	61	20	0.32	5.784	0.032	0.07280	0.00105	1028	5	1008	29	-2.0
S	80	80	156	284	1.83	5.942	0.030	0.07326	0.00065	1003	5	1021	18	1.8
S	71	71	111	154	1.39	5.817	0.029	0.07340	0.00075	1023	5	1025	21	0.2
S	120	120	194	86	0.44	5.669	0.051	0.07350	0.00070	1047	9	1028	19	-1.9
S	16	16	50	59	1.17	5.336	0.037	0.07460	0.00115	1107	7	1058	31	-4.7
S	131	131	148	82	0.55	5.525	0.032	0.07463	0.00060	1072	6	1059	16	-1.3
S	88	88	393	224	0.57	5.394	0.022	0.07473	0.00050	1096	4	1061	13	-3.3
S	87	87	144	138	0.96	5.394	0.025	0.07480	0.00065	1096	5	1063	17	-3.1
S	141	141	87	98	1.12	5.698	0.029	0.07490	0.00075	1042	5	1066	20	2.2
S	52	52	350	141	0.40	5.402	0.022	0.07492	0.00055	1095	4	1066	15	-2.7

Table 1. continued

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	$^{238}U/^{206}Pb$	± 1σ	$^{207}Pb/^{206}Pb$	± 1σ	$^{238}U/^{206}Pb^*$	± 1σ	$^{207}Pb^*/^{206}Pb^*$	± 1σ	Disc. (%)
S	123	123	361	281	0.78	5.537	0.021	0.07519	0.00055	1070	4	1074	15	0.3
S	95	95	431	48	0.11	5.316	0.020	0.07564	0.00050	1111	4	1086	13	-2.4
S	135	135	651	100	0.15	5.426	0.021	0.07567	0.00050	1090	4	1086	13	-0.4
S	116	116	96	105	1.09	5.336	0.027	0.07570	0.00085	1107	5	1087	23	-1.9
S	42	42	282	171	0.61	5.249	0.025	0.07574	0.00055	1124	5	1088	15	-3.3
S	146	146	46	59	1.29	5.359	0.043	0.07580	0.00105	1103	8	1090	28	-1.2
S	82	82	251	47	0.19	5.336	0.023	0.07585	0.00060	1107	4	1091	16	-1.5
S	117	117	171	132	0.77	5.672	0.027	0.07586	0.00060	1047	5	1091	16	4.1
S	28	28	286	173	0.61	5.214	0.024	0.07587	0.00055	1131	5	1092	15	-3.6
S	147	147	260	144	0.55	5.467	0.022	0.07638	0.00065	1083	4	1105	17	2.0
S	70	70	105	157	1.50	5.388	0.036	0.07640	0.00095	1097	7	1106	25	0.7
S	145	145	191	209	1.09	5.325	0.026	0.07680	0.00070	1109	5	1116	18	0.6
S	89	89	30	64	2.13	5.035	0.034	0.07710	0.00115	1168	7	1124	30	-3.9
S	63	63	181	178	0.98	5.010	0.023	0.07744	0.00065	1173	5	1133	17	-3.6
S	109	109	37	49	1.32	5.285	0.034	0.07770	0.00105	1117	7	1139	27	1.9
S	125	125	123	288	2.34	5.102	0.030	0.07905	0.00065	1154	6	1173	16	1.7
S	64	64	286	100	0.35	4.482	0.018	0.08278	0.00060	1298	5	1264	14	-2.7
S	105	105	76	1	0.01	4.669	0.021	0.08320	0.00080	1251	5	1274	19	1.8
S	13	13	240	55	0.23	4.658	0.025	0.08368	0.00070	1254	6	1285	16	2.4
S	4	4	368	352	0.96	4.440	0.018	0.08409	0.00055	1309	5	1295	13	-1.1
S	81	81	113	122	1.08	4.367	0.021	0.08430	0.00085	1329	6	1299	20	-2.3
S	114	114	164	158	0.96	4.419	0.020	0.08430	0.00070	1315	5	1299	16	-1.2
S	111	111	148	132	0.90	4.419	0.018	0.08467	0.00065	1315	5	1308	15	-0.5
S	93	93	83	114	1.38	4.529	0.019	0.08470	0.00080	1286	5	1309	18	1.7
S	74	74	111	123	1.11	4.266	0.017	0.08630	0.00075	1358	5	1345	17	-0.9
S	36	36	318	224	0.70	4.239	0.019	0.08978	0.00065	1365	5	1421	14	3.9
S	19	19	280	189	0.68	4.067	0.015	0.08982	0.00060	1417	5	1422	13	0.3
S	137	137	83	69	0.84	3.975	0.043	0.09100	0.00090	1447	14	1447	19	0.0
S	38	38	63	106	1.67	3.830	0.020	0.09240	0.00100	1495	7	1476	21	-1.3
S	83	83	199	122	0.61	4.024	0.043	0.09370	0.00085	1431	14	1502	17	4.7
S	130	130	70	116	1.65	3.552	0.021	0.09480	0.00075	1599	8	1524	15	-4.9
S	22	22	49	19	0.39	3.659	0.019	0.09630	0.00115	1558	7	1554	22	-0.3
S	104	104	157	116	0.74	3.639	0.017	0.09919	0.00075	1565	7	1609	14	2.7
S	59	59	177	135	0.76	3.631	0.015	0.09939	0.00075	1568	6	1613	14	2.8
S	7	7	191	121	0.64	3.475	0.014	0.10012	0.00075	1631	6	1626	14	-0.3
S	3	3	98	87	0.89	3.632	0.026	0.10140	0.00085	1568	10	1650	16	5.0
S	53	53	208	156	0.75	3.308	0.013	0.10183	0.00070	1703	6	1658	13	-2.7
S	29	29	337	199	0.59	3.364	0.012	0.10249	0.00070	1678	5	1670	13	-0.5

Table 1. continued

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$\frac{^{232}Th}{^{238}U}$	$^{238}U/^{206}Pb$	± 1σ	$^{207}Pb/^{206}Pb$	± 1σ	$^{238}U/^{206}Pb^*$	± 1σ	$^{207}Pb^*/^{206}Pb^*$	± 1σ	Disc. (%)
S	67	67	214	173	0.81	3.365	0.014	0.10259	0.00070	1677	6	1672	13	-0.4
S	136	136	164	93	0.57	3.360	0.015	0.10260	0.00080	1679	7	1672	14	-0.5
S	65	65	261	150	0.57	3.297	0.014	0.10448	0.00070	1708	6	1705	12	-0.1
S	142	142	147	97	0.66	3.248	0.014	0.10551	0.00080	1730	6	1723	14	-0.4
S	24	24	271	89	0.33	3.107	0.013	0.10744	0.00070	1799	7	1756	12	-2.4
S	55	55	174	94	0.54	3.085	0.015	0.10760	0.00080	1810	8	1759	14	-2.9
S	107	107	71	62	0.87	3.018	0.013	0.10890	0.00090	1845	7	1781	15	-3.6
S	1	1	298	79	0.27	3.240	0.013	0.10922	0.00075	1734	6	1786	13	2.9
S	62	62	220	160	0.72	2.581	0.009	0.12578	0.00085	2111	7	2040	12	-3.5
S	77	77	122	123	1.01	2.799	0.011	0.12600	0.00090	1969	7	2043	13	3.6
S	32	32	144	110	0.76	2.118	0.009	0.15794	0.00105	2493	9	2434	11	-2.4
S	150	150	224	149	0.67	2.054	0.008	0.16623	0.00110	2557	8	2520	11	-1.5
S	86	86	200	222	1.11	2.030	0.009	0.18270	0.00120	2582	9	2678	11	3.6
S	15	15	127	73	0.57	1.809	0.007	0.20110	0.00130	2837	9	2835	11	-0.1
S	76	76	361	67	0.19	1.742	0.010	0.22901	0.00140	2925	14	3045	10	4.0
S	144	144	91	228	2.51	1.639	0.008	0.23370	0.00155	3070	12	3078	11	0.2
S	61	61	208	102	0.49	1.611	0.007	0.24670	0.00160	3113	11	3164	10	1.6
S	110	110	64	64	1.00	1.585	0.007	0.25080	0.00170	3154	12	3190	11	1.1
S	20	20	319	104	0.33	1.562	0.006	0.26745	0.00165	3189	10	3291	10	3.1
S	140	140	80	66	0.82	1.462	0.007	0.26960	0.00175	3359	13	3304	10	-1.7
D	41	41	71	71	1.00	5.274	0.031	0.07380	0.00090	1119	6	1036	25	-8.0
D	30	30	96	137	1.43	5.195	0.030	0.07410	0.00085	1135	6	1044	23	-8.7
D	124	124	54	83	1.53	5.233	0.027	0.07470	0.00100	1127	5	1060	27	-6.3
D	69	69	117	82	0.70	5.784	0.032	0.07580	0.00070	1028	5	1090	19	5.7
D	56	56	99	72	0.73	4.912	0.021	0.07630	0.00080	1195	5	1103	21	-8.3
D	23	23	160	30	0.19	5.099	0.114	0.08430	0.00080	1154	24	1299	18	11.2
D	14	14	383	79	0.21	5.035	0.024	0.08460	0.00080	1168	5	1306	18	10.6
D	17	17	149	81	0.54	4.776	0.055	0.08500	0.00140	1226	13	1316	32	6.8
D	119	119	151	135	0.89	4.822	0.026	0.08590	0.00070	1215	6	1336	16	9.1
D	139	139	199	93	0.47	4.248	0.021	0.09220	0.00075	1363	6	1472	15	7.4
D	78	78	338	214	0.63	4.270	0.036	0.09221	0.00065	1356	10	1472	13	7.8
D	58	58	488	110	0.23	3.971	0.065	0.10120	0.00125	1448	21	1646	23	12.1
D	45	45	316	135	0.43	3.658	0.038	0.10267	0.00065	1558	14	1673	12	6.9
D	57	57	272	103	0.38	3.503	0.055	0.10665	0.00075	1619	22	1743	13	7.1
D	39	39	174	258	1.48	4.435	0.062	0.10680	0.00095	1311	17	1746	16	24.9
D	34	34	615	355	0.58	3.092	0.036	0.14976	0.00100	1806	18	2343	11	22.9
D	127	127	205	88	0.43	2.275	0.010	0.16157	0.00105	2348	9	2472	11	5.0

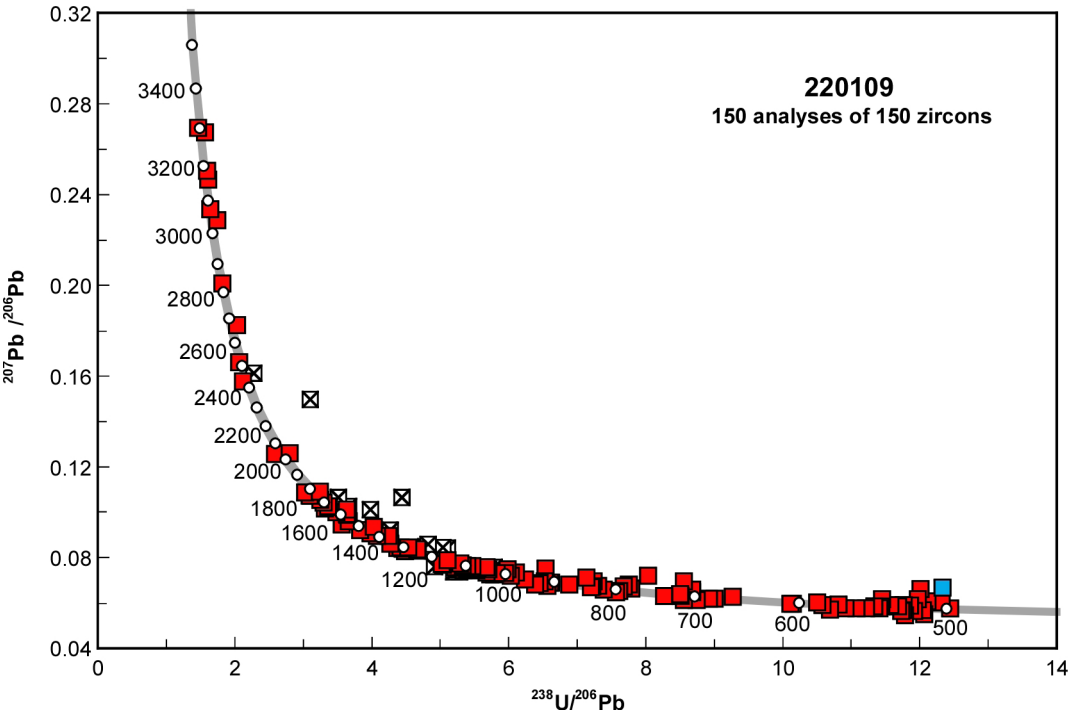

Figure 3. U–Pb analytical data, not corrected for common Pb, for sample 220109: sandstone, Wendy 1. Blue square indicates Group Y (youngest detrital zircon); red squares indicate Group S (older detrital zircons); crossed squares indicate Group D (date >1000 Ma and discordance >5%)

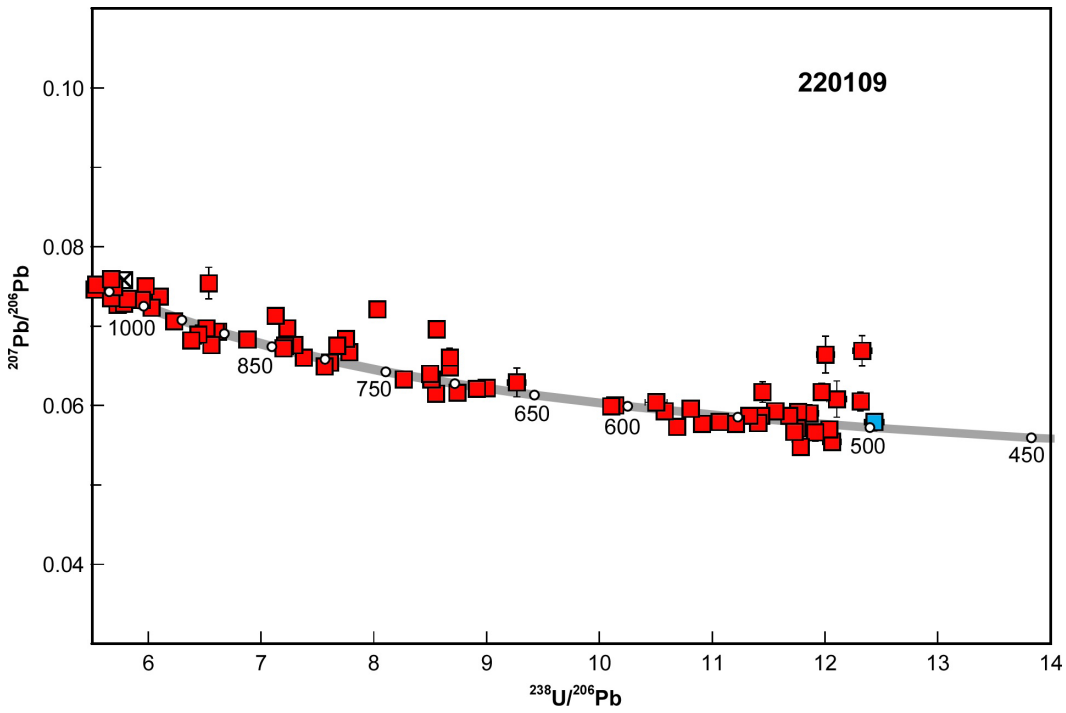

Figure 4. Expanded view of U–Pb analytical data, not corrected for common Pb, for sample 220109: sandstone, Wendy 1. Symbols as in Figure 3

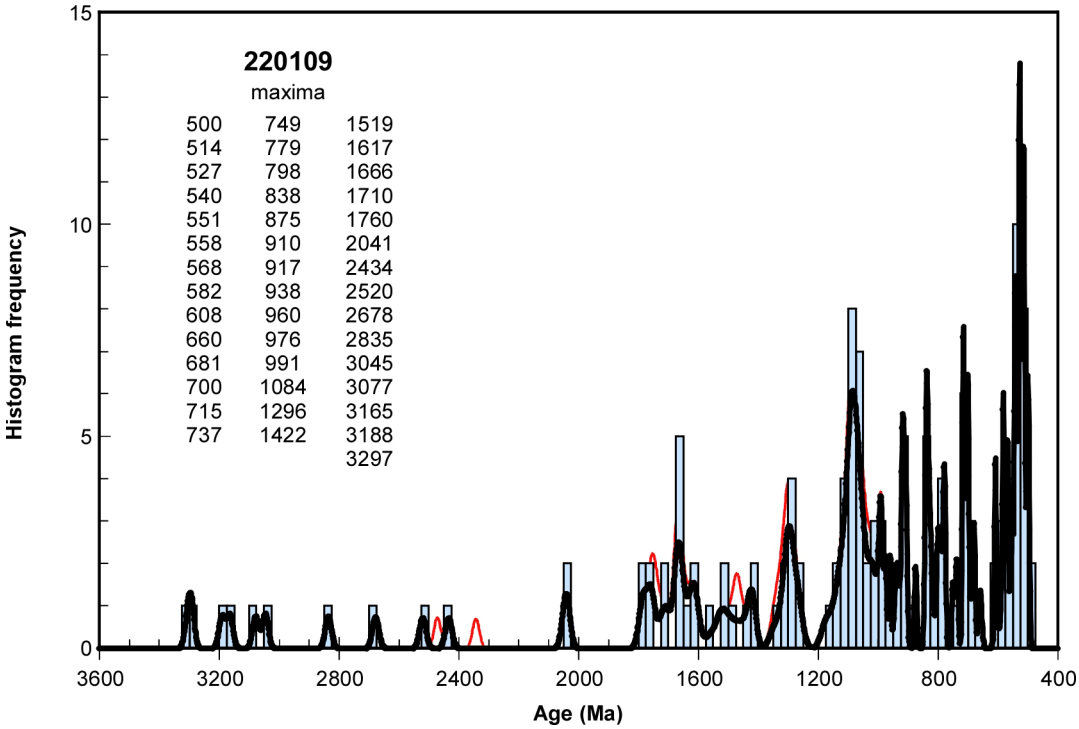

Figure 5. Probability density diagram and histogram for sample 220109: sandstone, Wendy 1. Thick curve, maxima values, and frequency histogram (bin width 25 Ma) only include accepted data (133 analyses of 133 zircons). Thin curve includes all data (150 analyses of 150 zircons)

Interpretation

The analyses are concordant to strongly discordant (Figs 3, 4). Seventeen analyses >1000 Ma are >5% discordant. The dates obtained from these 17 analyses (Group D; Table 1) are unreliable, and considered not to be geologically significant. The remaining 133 analyses can be divided into two groups, based on their $^{207}Pb/^{206}Pb$ and $^{238}U/^{206}Pb^*$ ratios.

Group Y comprises one analysis (Table 1), which yields a 207-corrected $^{238}U/^{206}Pb^*$ date of 497 ± 4 Ma (1σ).

Group S comprises 132 analyses (Table 1), which yield $^{207}Pb/^{206}Pb$ or $^{238}U/^{206}Pb^*$ dates of 3304–498 Ma.

It is possible that all of the analyses are of unmodified detrital zircons, in which case the date of 497 ± 4 Ma (1σ) for the single analysis in Group Y represents a maximum depositional age for the sandstone. A more conservative estimate of the maximum depositional age can be based on the weighted mean $^{238}U/^{206}Pb^*$ date of 501 ± 8 Ma (MSWD = 2.3) for the four youngest analyses in Groups Y and S.

The data for combined Groups Y and S indicate significant age components at c. 1760, 1666, 1617, 1519, 1422, 1296, 1084, 991, 976, 917, 910, 838, 779, 715, 700, 582, 540, 527, 514, and 500 Ma, based on contributions from approximately 3, 5, 3, 3, 3, 7, 19, 7, 3, 4, 4, 4, 3, 3, 3, 3, 4, 5, 7, and 3 analyses, respectively (Fig. 5). These are interpreted as the ages of zircon-crystallizing rocks in the detrital source region(s), or as the ages of detrital components within sediments that have been reworked into this rock.

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